Ch-5- Rev- Problems- Summer 26

5.1 The coefficients of a digital Lattice filter are $K_1 = -0.266$ and $K_2 = 0.69$.
(a) Draw the realization diagram for the Lattice filter.
(b) Calculate the impulse response coefficients for the filter.
(c) Draw its equivalent transversal FIR filter's structure.

5.2 The coefficients of transversal filter are $h_0 = h_1 = -0.266$ and $h_2 = 0.69$.
(a) Draw the realization structure for this the transversal filter.
(b) Find the difference equation for this filter.
(c) Find its transfer function, H(Z).
(d) Draw its equivalent Lattice structure.
(e) Calculate the values of the coefficients for the lattice structure.

5.3 The transfer function for an FIR filter is given by: $H(z) = 1 - 1.3435z^{-1} + 0.9025z^{-2}$
Draw the realization block diagram for each of the following cases:
1- Transversal structure;
2- A two-stage lattice structure.

Calculate the values of the coefficients for the lattice structure.